Plasmid preparation by alkaline lysis

Under alkaline conditions, high molecular weight DNA is selectively denatured while covalently closed circular plasmid DNA remains double-stranded. When the solution is rapidly neutralized, denatured chromosomal DNA forms an insoluble clot, while plasmid DNA remains in solution (Birnboim and Doly, 1979).

This method yields tens of micrograms of highly pure plasmid DNA from mini-preps (a few milliliters of culture), and up to milligrams from a liter-scale culture.

The plasmids are purified over a silica matrix (*Alternative A*) or by alcohol precipitation (*Alternative B*).

Risk assessment

▷ Wear gloves, safety glasses, lab coat

Reviewed: May 21, 2026

Procedures

›› *Alkaline lysis*

- ☐ LB medium (Miller) R0030
- ☐ Buffer P1 (Resuspension buffer), pH 8.0 R0006
- ☐ Buffer P2 (Lysis buffer) R0007
- ☐ Buffer N3 (Neutralization buffer), pH 4.8 R0009

(1.) Grow bacteria in 1–5 mL LB (Miller) overnight to stationary phase. Collect the cells by centrifugation.

> ***Hint:*** For culture tubes, spin at 3 000 × *g* for 5 min. For 2 mL centrifuge tubes, a brief spin at 11 000 × *g* is sufficient.

(2.) Resuspend the pellet in 150 µL Buffer P1.

> ***Hint:*** Buffers P1, P2, N3, PB, and PE refer to standard silica-column miniprep kit components (e.g., QIAGEN QIAprep). Equivalent buffers from other suppliers can be substituted.

(3.) Add 250 µL Buffer P2. Mix by gently inverting the tube 6–8 times. Incubate at room temperature for no more than 5 min or until the lysate appears clear.

> ***Critical:*** Do not vortex or pipette to avoid shearing chromosomal DNA.

(4.) Add 350 µL Buffer N3. Mix gently by inversion 6–8 times.

> ***Critical:*** Ensure complete neutralization to precipitate protein and chromosomal DNA. If Buffer P2 contains thymolphthalein, mix until the solution is colorless.

(5.) Centrifuge at 17 000 × *g* for 3 min to pellet precipitated material. Repeat if the lysate remains turbid.

(6.) Transfer the clear supernatant to a fresh tube for purification. Avoid any carryover of precipitate.

A › *Purification of plasmid DNA over a silica matrix*

- ☐ Buffer PB (Binding buffer), pH 4.8 R0010
- ☐ Buffer PE (Wash buffer), pH 7.4 R0011
- ☐ 5 mM Tris hydrochloride, pH 8.0 (R0056 for 1 M)
- ☐ Spin column (1 / 1 rxn)

(1.) Apply up to 700 µL of supernatant to a silica spin column. Centrifuge at 11 000 × *g* for 1 min, or use a vacuum manifold.

> ***Hint:*** Silica membranes typically bind 15–25 µg DNA. They may be reused to purify more of the same DNA.

(2.) *Optional:* Wash the silica matrix with 450 µL Buffer PB.

> ***Hint:*** Recommended if plasmids are isolated from strains containing high endonuclease activity such as HB101 or JM109.

(3.) Wash with 650 µL Buffer PE.

(4.) Dry the silica matrix by centrifugation at 11 000 × *g* for 2 min.

(5.) Place the spin column into a clean microcentrifuge tube.

(6.) Apply 25–55 µL 5 mM Tris pH 8.0 to the membrane. Incubate for 1 min, then elute the plasmid by centrifugation at 11 000 × *g* for 1 min.

> ***Hint:*** For higher yield or large plasmids, warm elution buffer to 70 °C.

(7.) *Optional:* Repeat elution using the same or half the volume to improve yield. Pool eluates if desired.

B › *Purification by alcohol precipitation*

- ☐ Isopropyl alcohol
  - *or*: 100% Ethanol
- ☐ 3 M Sodium acetate, pH 5.2 (*optional*) R0045
- ☐ Buffer PE (Wash buffer) R0011
- ☐ 5 mM Tris hydrochloride, pH 8.0 (R0056 for 1 M)

(1.) Transfer up to 700 µL of supernatant to a clean microcentrifuge tube.

(2.) Add an equal volume of isopropyl alcohol, or 2 vol of ethanol, to precipitate the DNA. Vortex to mix thoroughly.

(3.) Centrifuge at 11 000 × *g* for 5 min. Discard the supernatant and briefly spin to remove residual solvent. The pellet will appear nearly transparent.

(4.) Wash with 650 µL Buffer PE or freshly prepared 70% ethanol without disturbing the pellet.

(5.) Air-dry the pellet for 2–5 min, or dry in a vacuum centrifuge for 1 min.

(6.) Resuspend in 25–55 µL 5 mM Tris pH 8.0 or water.

C › *Purification of plasmid DNA by anion-exchange chromatography*

- ☐ Buffer P3 (Neutralization buffer), pH 5.5 R0008
- ☐ Buffer QBT (Equilibration buffer), pH 7.0 R0197
- ☐ Buffer QC (Wash buffer), pH 7.0 R0198
- ☐ Buffer QF (Elution buffer), pH 8.5 R0199
- ☐ Anion-exchange column (1 / 1 rxn)

(1.) Equilibrate the anion-exchange column with the recommended volume of Buffer QBT and allow it to drain by gravity flow.

> ***Hint:*** Buffer and lysate volumes scale with tip size: roughly 4 mL per step for a 100 µg tip, 10 mL for a 500 µg tip, and 35 mL for a 2 500 µg tip.

(2.) Apply the P3-neutralized, cleared lysate to the equilibrated column and let it pass through by gravity.

> ***Critical:*** The column must not run dry between steps; if flow stalls, refill with Buffer QC and continue.

(3.) Wash the column twice with the recommended volume of Buffer QC.

(4.) Elute the plasmid DNA with the recommended volume of Buffer QF into a clean tube.

(5.) Precipitate the DNA by adding 0.7 vol isopropanol at room temperature, wash, and resuspend as described in *Alternative B*.

Analyses

- ○ Measure DNA concentration and purity by UV absorbance at 260 nm, 280 nm, and 230 nm.

Nucleic acid	A260 = 1.0	A260/A280	A260/A230
dsDNA	50 ng/µL	1.8–1.9	2.0–2.2

> ***Quality assurance:*** Extinction coefficients vary with nucleotide composition. A260/A280 ratios increase with pH and vary by 0.2–0.3 between instruments. RNA contamination elevates the ratio; protein or phenol contamination depresses it.

- ○ Confirm plasmid identity by restriction digestion SOP 0033 and agarose gel electrophoresis SOP 0032.

> ***Hint:*** Use one or more enzymes predicted to linearize the plasmid or release a diagnostic fragment.

Materials

Alkaline lysis

LB medium (Miller) R0030

Buffer P1 (Resuspension buffer) R0006, pH 8.0, Store at 4 °C.

Buffer P2 (Lysis buffer) R0007

Buffer N3 (Neutralization buffer) R0009, pH 4.8, Store at room temperature.

Purification of plasmid DNA over a silica matrix

Buffer PB (Binding buffer) R0010, pH 4.8

Buffer PE (Wash buffer) R0011, pH 7.4

5 mM Tris hydrochloride (R0056 for 1 M), pH 8.0

Spin column (1 / 1 rxn), silica, 15–25 µg dsDNA capacity

- **QIAprep spin column**, QIAGEN, QIAprep Spin Miniprep Kit, 27104

Purification by alcohol precipitation

Precipitating alcohol

- **Isopropyl alcohol**, CAS 67-63-0
- *or* **Ethanol**, CAS 64-17-5, 100%

Sodium acetate R0045 (*optional*), 3 M, pH 5.2

Buffer PE (Wash buffer) R0011

5 mM Tris hydrochloride (R0056 for 1 M), pH 8.0

Purification of plasmid DNA by anion-exchange chromatography

Buffer P3 (Neutralization buffer) R0008, pH 5.5, Store at 4 °C.

Buffer QBT (Equilibration buffer) R0197, pH 7.0, Store at room temperature.

Buffer QC (Wash buffer) R0198, pH 7.0, Store at room temperature.

Buffer QF (Elution buffer) R0199, pH 8.5, Store at room temperature.

Anion-exchange column (1 / 1 rxn), silica-based, hydrophilic anion-exchange resin, gravity-flow tip; capacity scaled to plasmid yield (100 µg, 500 µg, 2 500 µg)

- **QIAGEN-tip**, QIAGEN, QIAGEN Plasmid Midi/Maxi/Mega Kit

Recipes

Buffer P1 (Resuspension buffer), pH 8.0

Amount	Ingredient		Stock	Final
12.5 mL	Tris hydrochloride (Tris-Cl), pH 8.0 R0056		1 M	50 mM
5.0 mL	EDTA, pH 8.0	R0017	0.5 M	10 mM
250 µL	RNase A, bovine pancreas R0040 Respiratory sensitization (H334)	[9001-99-4]	7 U/µL	7 U/mL
To 250 mL	Water, reagent-grade			

Store at 4 °C. ***Critical:*** If the buffer is older than six months, supplement with fresh RNase A.

Buffer P1 (Resuspension buffer)

50 mM Tris-Cl, 10 mM EDTA, 7 U/mL RNase A, pH 8.0

No GHS hazards at the indicated concentrations

- ☐ Hold for EHS review before disposal

Date: Sign: R0006

Buffer P2 (Lysis buffer)

Amount	Ingredient		Stock	Final
2.0 g	NaOH Skin corrosion (H314)	[1310-73-2]	40.00 g/mol	200 mM
12.5 mL	SDS Skin irritation (H315); Serious eye damage (H318)	R0047	20%	1%
125 mg	Thymolphthalein	☐ [125-20-2]	430.54 g/mol	0.05%
To 250 mL	Water, reagent-grade			

This is why: Thymolphthalein serves as pH indicator to monitor the neutralization reaction with N3; it is optional.

Buffer P2 (Lysis buffer)

200 mM NaOH, 1% SDS, ☐ 0.05% Thymolphthalein

WARNING

Skin and serious eye irritation

- ☐ Collect as HAZARDOUS WASTE
- ☐ DO NOT wash into sewer

Date: Sign: R0007

Buffer N3 (Neutralization buffer), pH 4.8

Amount	Ingredient		Stock	Final
100.0 g	Guanidine · HCl	[50-01-1]	95.53 g/mol	4.2 M
22.0 g	KOAc	[127-08-2]	98.14 g/mol	0.9 M
To 250 mL	Water, reagent-grade			

Adjust pH with acetic acid. Store at room temperature. ***Hint:*** For use with silica-gel membranes as in the QIAGEN QIAprep Spin Miniprep Kit.

Buffer N3 (Neutralization buffer)

4.2 M Guanidine, 0.9 M KOAc, pH 4.8

WARNING

Skin and serious eye irritation; Acute oral toxicity

- ☐ Neutralise to pH 6–9
- ☐ Flush to drain with excess water

Date: Sign: R0009

Buffer PB (Binding buffer), pH 4.8

Amount	Ingredient		Stock	Final
119.0 g	Guanidine · HCl	[50-01-1]	95.53 g/mol	5.0 M
75 mL	Isopropyl alcohol Flammable liquid (H225)	[67-63-0]	100%	30%
To 250 mL	Water, reagent-grade			

Adjust pH with acetic acid before adding isopropyl alcohol. Store at room temperature.

Buffer PB (Binding buffer)

5.0 M Guanidine, 30% Isopropyl alcohol, pH 4.8

DANGER

Flammable liquid; Skin and serious eye irritation; Drowsiness or dizziness; Acute oral toxicity

- ☐ Collect as ORGANIC SOLVENT WASTE
- ☐ Keep away from oxidizers

Date: Sign: R0010

Buffer PE (Wash buffer), pH 7.4

Amount	Ingredient		Stock	Final
2.5 mL	Tris-Cl, pH 7.4	R0055	1 M	10 mM
200 mL	Ethanol, reagent-grade Flammable liquid (H225)	[64-17-5]	100%	80%
To 250 mL	Water, reagent-grade			

Store at room temperature.

Buffer PE (Wash buffer)

10 mM Tris-Cl, 80% Ethanol, pH 7.4

DANGER

Flammable liquid

- ☐ Collect as ORGANIC SOLVENT WASTE
- ☐ Keep away from oxidizers

Date: Sign: R0011

Buffer P3 (Neutralization buffer), pH 5.5

Amount	Ingredient		Stock	Final
73.0 g	KOAc	[127-08-2]	98.14 g/mol	2.9 M
To 250 mL	Water, reagent-grade			

Adjust pH with acetic acid. Store at 4 °C. ***Hint:*** For use with anion-exchange columns as in the QIAGEN Plasmid Kit.

Buffer P3 (Neutralization buffer)

pH 5.5

EPA Safer Choice low-concern

- ☐ Flush to drain with excess water

Date: Sign: R0008

Buffer QBT (Equilibration buffer), pH 7.0

Amount	Ingredient		Stock	Final
10.95 g	NaCl	[7647-14-5]	58.44 g/mol	750 mM
2.62 g	MOPS	[1132-61-2]	209.26 g/mol	50 mM
37.5 mL	Isopropyl alcohol Flammable liquid (H225)	[67-63-0]	100%	15%
375 µL	Triton™ X-100 Serious eye damage (H318)	[9036-19-5]	100%	0.15%
To 250 mL	Water, reagent-grade			

Adjust pH with NaOH before adding isopropanol. Store at room temperature.

Buffer QBT (Equilibration buffer)

750 mM NaCl, 50 mM MOPS, 15% Isopropyl alcohol, 0.15% Triton™ X-100, pH 7.0

DANGER

Flammable liquid; Serious eye irritation; Drowsiness or dizziness

- ☐ Collect as ORGANIC SOLVENT WASTE
- ☐ Keep away from oxidizers

Date: Sign: R0197

Buffer QC (Wash buffer), pH 7.0

Amount	Ingredient		Stock	Final
2.62 g	MOPS	[1132-61-2]	209.26 g/mol	50 mM
14.61 g	NaCl	[7647-14-5]	58.44 g/mol	1.0 M
37.5 mL	Isopropyl alcohol Flammable liquid (H225)	[67-63-0]	100%	15%
To 250 mL	Water, reagent-grade			

Adjust pH with NaOH before adding isopropanol. Store at room temperature.

Buffer QC (Wash buffer)

50 mM MOPS, 1.0 M NaCl, 15% Isopropyl alcohol, pH 7.0

DANGER

Flammable liquid; Serious eye irritation; Drowsiness or dizziness

- ☐ Collect as ORGANIC SOLVENT WASTE
- ☐ Keep away from oxidizers

Date: Sign: R0198

Buffer QF (Elution buffer), pH 8.5

Amount	Ingredient		Stock	Final
12.5 mL	Tris hydrochloride (Tris-Cl), pH 8.5 R0055		1 M	50 mM
18.26 g	NaCl	[7647-14-5]	58.44 g/mol	1.25 M
37.5 mL	Isopropyl alcohol Flammable liquid (H225)	[67-63-0]	100%	15%
To 250 mL	Water, reagent-grade			

Store at room temperature.

Buffer QF (Elution buffer)

50 mM Tris-Cl, 1.25 M NaCl, 15% Isopropyl alcohol, pH 8.5

DANGER

Flammable liquid; Serious eye irritation; Drowsiness or dizziness

- ☐ Collect as ORGANIC SOLVENT WASTE
- ☐ Keep away from oxidizers

Date: Sign: R0199

Troubleshooting

Alkaline lysis

In Step 4:

- Chromosomal DNA does not form a white precipitate but remains viscous.
  - Reduce culture volume and repeat the preparation with gentler handling.

In Step 6:

- Chromosomal DNA in the eluate
  - Use cultures within 12–16 h post-inoculation. Older cultures release degraded chromosomal DNA. If not processing immediately, store the bacterial pellet at −20 °C.
  - Mix lysate gently after adding Buffer P2. Limit lysis to 5 min to avoid DNA shearing.
  - After adding Buffer N3, mix immediately and gently to allow complete neutralization.
- RNA contamination in eluate
  - Verify RNase A was added to Buffer P1 before use. Supplement fresh RNase if needed.

Purification of plasmid DNA over a silica matrix

In Step 6:

- Little or no DNA in eluate
  - Ensure elution buffer is low salt and within pH 7.0 and pH 8.5.

Purification by alcohol precipitation

In Step 3:

- No visible pellet after isopropanol precipitation
  - Centrifuge for longer (10 min µN t). Mark the tube orientation before spinning to locate it on the expected side.
  - Add 0.1 vol of 3 M sodium acetate pH 5.2before adding isopropanol to improve precipitation of dilute samples.

Purification of plasmid DNA by anion-exchange chromatography

In Step 4:

- Low yield after elution
  - Pre-warm Buffer QF to 50 °C for large plasmids (>10 kb) to improve recovery.

List of references

H. Birnboim and J. Doly, *Nucleic Acids Res.* **7**(6), 1513—1523 (1979).

Change log

2016-03-15	Rahul Patharkar	Alkaline lysis with ethanol precipitation; from the Patharkar life-science protocols hub (patharkar.com).
2023-05-30	Benjamin C. Buchmuller	Adaptation as SOP, supplemented from Qiagen "QIAprep Spin Miniprep Kit" handbook.
2026-05-21	Benjamin C. Buchmuller	Fix two transcription errors in Buffer P1: RNase A final concentration (was "7 enzymeunit per liter", now "7 enzymeunit per milliliter"; stock and amount unchanged, prep unaffected) and EDTA amount (was 10.0 mL, now 5.0 mL — recipe v1 over-prepared at 20 mM; corrected recipe yields the intended 10 mM, matching the Qiagen reference).